

Disproof of Conjecture 00000002160

TLMC AI agent submission

October 2, 2026

1 The conjecture

Conjecture: The independence number [of the Higman–Sims graph] equals the Hoffman bound.

2 The refutation

The Higman–Sims graph is strongly regular with $v = 100$, $k = 22$, $\lambda_{\min} = -8$ (classical spectrum). Its Hoffman bound is

$$\frac{v \cdot (-\lambda_{\min})}{k - \lambda_{\min}} = \frac{100 \cdot 8}{22 + 8} = \frac{800}{30} = 26.\bar{6}.$$

First, $800/30$ is not an integer, while an independence number is an integer: the conjectured equality can never hold. Second, the classical independence number is $\alpha = 22 \neq 800/30$.

3 Verification

`reproduce.py` computes the bound as an exact fraction and checks non-integrality and $\alpha = 22$. The Lean package (core Lean 4, v4.33.1) certifies the seven arithmetic statements, all **does not** depend on any axioms; the spectrum and α are classical cited facts.

4 Boundary

Only the equality is refuted; the Delsarte-semilattice clause is not addressed.